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Seeing red: pteridine-based colour and male quality in a dragon lizard

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The animal kingdom is represented by a spectacular diversity of colourful species. However, for particular groups, such as lizards, we still know little about the nature of these colours and what they might signal about individual quality. In eastern water dragons (*Intellagama lesueurii*), males are larger than females and have a red venter. We first identified the pigments responsible for the red coloration. We then used visual modelling to determine how conspicuous red colouration is to a dragon lizard. Finally, we asked whether colour signalled male morphology and whether parasitized males had reduced coloration. The red coloration of male water dragons is produced by both carotenoid and pteridine classes of pigments, but predominantly by drosopterin, a coloured pteridine. Heavy males with large heads and in good body condition and those with fewer ticks had less intense red coloration on the chest, abdomen and thighs. However, size was not correlated with any colour variables. We explain the negative relationship between colour conspicuousness and morphological variables by the fact that less conspicuous males might have more melanin interspersed with red. Future studies are needed to investigate the interplay between pteridines and melanin in social signals of male quality in dragon lizards.

ADDITIONAL KEYWORDS: colour – honest signalling – lizard – male–male competition – parasites – reptile.

INTRODUCTION

Understanding the evolution and maintenance of phenotypic diversity is a major focus in evolutionary biology. To this end, the study of animal coloration has grown explosively in the past few decades because colour signals are strongly influenced by a wide range of environmental and social factors that drive divergence in colour signals (Cuthill *et al.*, 2017). In order to understand how this diversity in animal colour signals is maintained, we also need to understand the proximate mechanisms governing colour production (McLean *et al.*, 2017). Coloration in animals is produced either by nanostructures (i.e. structural colour) or by pigments (Cuthill *et al.*, 2017). In the case of pigments, these can either be sourced from the environment or produced *de novo*. Understanding the relative ratios of these pigments is useful because it helps us understand constraints to signal production and whether a colour signal is an indicator of the quality of an individual.

In reptiles, colour signals are a product of several proximate layers of colour pigments known collectively as the dermal chromatophore unit (Cooper & Greenberg, 1992). Chromatophores are pigment cells that are classed by colour (e.g. xanthophores and erythrophores are yellow to red) and pigment type (e.g. melanophores contain melanin). Two common chromatophores, xanthophores and erythrophores, contain pteridines and carotenoids, which filter light in a similar way, therefore producing similar colours (from yellow to red). The dermal chromatophore unit refers to the arrangement of chromatophores in the

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dermis (typically three cell layers): xanthophores/erythrophores compose the outermost layer, followed by iridophores, and the third layer is composed of melanophores (light absorbing; Bagnara & Matsumoto, 2006). The interaction of these three layers produces the colour seen in the skin. Therefore, differences in the presence and/or quantity of pteridines, carotenoids and melanins, and in the internal structure of iridophores and melanophores, can result in variation in colour properties (San-Jose *et al.*, 2013; Merkling *et al.*, 2015; Teyssier *et al.*, 2015).

Carotenoids and pteridines are the two main pigment types responsible for conspicuous coloration in reptiles (Steffen & McGraw, 2007; Kikuchi & Pfennig, 2012; Weiss *et al.*, 2012; Merkling *et al.*, 2015; McLean *et al.*, 2017). In addition to colour production, carotenoid pigments play a role in immune function as antioxidants and immunoregulators (Svensson & Wong, 2011), and they can be obtained only from an animal's environment (i.e. they cannot be synthesized *de novo*; Olson & Owens, 1998). Carotenoid-based colour is therefore typically viewed as an honest signal of the quality of an individual, because of the trade-offs between coloration and physiological function. Conversely, animals can synthesize pteridines from guanine triphosphate. Pteridines can act as cofactors in enzymatic reactions, and some forms may reduce oxidative stress or play a role in the immune system (Huber *et al.*, 1984; Oettl & Reibnegger, 2002). However, there is very little information about the potential trade-offs between the allocation of coloured pteridine synthesis and specific functions, such as immunity and the antioxidant system (Weiss *et al.*, 2012).

The association between immune function and colour production can also be apparent when considering parasite load; animals with more parasites may have a reduced colour signal. For example, satin bowerbirds (*Ptilonorhynchus violaceus*) infected with blood parasites have noticeably darker plumage than uninfected individuals (Doucet & Montgomerie, 2003). Lizards frequently have to deal with blood-, intestinal-, and ecto-parasites. Trying to understand the collective effect of these parasites is particularly challenging and takes carefully designed experimental studies. Nevertheless, it can be useful to count ectoparasites, because the number of ticks may be indicators of parasite loads and health (Bull & Burzacott, 1993; Olsson *et al.*, 2000; Megía-Palma *et al.*, 2017). In the lizard *Lacerta schreiberi*, higher parasite loads negatively impacted the expression of social colour signals (Megía-Palma *et al.*, 2017).

Sexual dichromatism, when the sexes differ in coloration, can evolve through sexual selection (but for exceptions, see Badyaev & Hill, 2003) either by acting as a signal of individual quality or because it is an arbitrary trait signalling attractiveness (Andersson, 1994). In lizards, extravagant male coloration may be accompanied by selection on head and body size such that larger males generally have an advantage in contests (Stuart-Fox & Ord, 2004). Colour traits can signal male quality in a number of ways: they may signal a singular aspect of male quality (thus adding to a general signal), they may reinforce the reliability of other traits, or they may aid in improving signal recognition and detection and in reducing receiver reaction times (Rowe, 1999; Martín & Lopez, 2009).

Lizards are good model systems for understanding the nature of colour signals because male coloration is frequently under sexual selection, and this is accompanied by selection on morphology, including body and head size (Cooper & Greenberg, 1992). Although male colour signals in lizards have been demonstrated to play a key role in contest competition (Whiting *et al.*, 2003) and alternative reproductive tactics (Sinervo & Lively, 1996), we know much less about the nature of this colour and how it is produced (but see McLean *et al.*, 2017).

Eastern water dragons (*Intellagama lesueurii*; Supporting Information, Fig. S1) are a suitable model system for investigating the nature of colour signals because males are significantly larger and heavier than females and have a red chest (Baird *et al.*, 2013). *Intellagama lesueurii* is the largest species of agamid lizard in Australia, with males capable of exceeding 300 mm snout–vent length (SVL) and weighing up to 1 kg (Thompson, 1993). They often retreat to water when threatened and either sleep submerged up to their necks or on branches overhanging water (Courtice, 1981; Thompson, 1993). In most males, colour starts from the throat region and continues along the venter to the vent; however, the extent of colour varies among males and sometimes extends down the forelimbs and thighs of some individuals. Males begin developing red on their ventral surface as juveniles, whereas females have mostly beige venters, although they may have some red too. As adults, coloration does not visibly change throughout the year (M. J. W., pers. obs.).

Adult males are particularly aggressive during the breeding season and adopt condition-dependent alternative reproductive tactics, in which resident males tend to be larger than satellite males (Baird *et al.*, 2012, 2014). In a high-density population, roughly equal numbers of males adopted each tactic (Baird *et al.*, 2012). These alternative reproductive tactics are plastic, and satellite males have been documented to usurp resident males or switch to a territorial tactic if a territory becomes available (Baird *et al.*, 2012). Interestingly, baseline testosterone and corticosterone levels are similar in both groups, probably because of the high density of males and the need to be ready either to defend a territory as a resident or to take over a territory as a satellite male (Baird *et al.*, 2014).

Male displays can consist of both head bobs and forelimb extensions (Cuervo & Shine, 2007). In the assessment phase, the throat and chest regions of males are clearly visible. However, males can also perform 'full shows', in which they laterally compress and raise their bodies while straightening and stiffening their limbs and tilting their heads upwards (Baird *et al.*, 2012), therefore making the coloration of the abdomen, legs and forelimbs also visible. Males are also known to charge opponents in contests by running bipedally, during which their leg coloration would be visible (Baird *et al.*, 2013). Male colour might thus be used in male–male contests as a signal, although its exact role remains to be established, as does the nature of this coloration.

We asked three questions. First, is male water dragon coloration carotenoid or pteridine based? Second, is the water dragon visual system able to discriminate the red ventral colour patch? Third, does colour signal male quality (as measured by body and head size, body mass and condition), and is colour intensity inversely correlated with parasite load?

MATERIAL AND METHODS

Study area

Lane Cove National Park is situated in a bushland valley in northern metropolitan Sydney, NSW, Australia, with Lane Cove River coursing through the middle of the valley. The vegetation in the park consists mainly of casuarina (*Casuarina glauca*) woodlands along the riverbanks, isolated patches of Sydney blue gum (*Eucalyptus saligna*) forest, or areas containing a combination of blackbutt (*Eucalyptus pilularis*), turpentine (*Syncarpia glomulifera*) and blue gum. The riverbank varies from densely vegetated to open, cleared areas consisting of wood chips and logs. We conducted fieldwork along 2 km of river in Lane Cove National Park (beginning, 33°47′29.54″S, 151°9′20.40″E; end, 33°47′12.84″S, 151°8′53.20″E) from September 2010 to January 2012. Males establish territories along the riverbank, which is typically in close proximity to walking trails or picnic areas; consequently, they are accustomed to humans and allow relatively close approach.

Carotenoid and pteridine pigment extraction

We extracted pigments from skin biopsies of red throat and dorsal brown body skin tissue from three mature male *I. lesueurii*. The brown body region was taken as an intended baseline, to determine the pigments generating the red throat coloration. We used the sequential carotenoid and pteridine extraction protocol developed for McLean *et al.* (2017), with only slight modifications. Samples were first dried and weighed, before being transferred to 1.5 mL microcentrifuge tubes (Eppendorf, Hamburg, Germany). To extract the carotenoids, 800 µL methanol:ethylacetate (6:4 v/v; MeOH, Thermo Fisher Scientific, Scoresby, Victoria, Australia; EtOAc, Sigma-Aldrich, St Louis, MO, USA) containing 0.01% butylated hydrotoluene (BHT) and a carotenoid internal standard (ISTD; 10 µM trans-β-apo-8′-carotenal; Sapphire Biosciences, Redfern, NSW, Australia) were added to each sample. Samples were homogenized using two 3 mm tungsten carbide beads on a TissueLyser II system (for two 5 min periods at 30 Hz, or until homogenized; Qiagen, Hilden, Germany). Homogenized samples were incubated for 30 min at 4 °C on an Eppendorf thermomixer at 1000 rpm and then centrifuged for 5 min at 4 °C at 12 000 rpm. The supernatants were collected and the pellets re-extracted for 30 min with an additional 400 µL of MeOH:EtOAc (6:4 v/v; containing BHT and ISTD) on the thermomixer. The sample was again centrifuged for 5 min at 4 °C at 12 000 rpm and the supernatant collected. The combined carotenoid fractions were then dried in a rotational vacuum concentrator (Speedvac), resuspended in 60 µL acetonitrile (ACN; containing 0.1% formic acid; Merck, Darmstadt, Germany):acetone (6.7:3.3 v/v; Sigma-Aldrich).

To extract pteridines, after the carotenoid extractions, 800 µL 2% ammonium hydroxide (NH_4OH; Sigma-Aldrich) containing an ISTD for pteridines (10 µM Biopterin-d3; Sapphire Biosciences) was added to the tissue pellets. Samples were incubated for 30 min at 4 °C on an Eppendorf thermomixer at 1000 rpm and then centrifuged for 5 min at 4 °C at 12 000 rpm. The supernatants were collected and the pellets re-extracted for 30 min with another 400 µL 2% NH_4OH (containing ISTD) on the thermomixer. The samples were again centrifuged for 5 min at 4 °C at 12 000 rpm and the supernatant collected. The combined pteridine fractions were then dried in a Speedvac, resuspended first in 60 µL 0.02% NH_4OH, then diluted with the addition of 240 µL ultra-pure water. Isoxanthopterin was analysed after a 100-fold dilution.

Liquid chromatography and mass spectrometry analysis

We quantified carotenoids and pteridines using liquid chromatography (LC) and mass spectrometry (MS) methods, modified from McLean *et al.* (2017), in two separate analyses on an Agilent 6490 triple quadrupole MS system with a Jet Stream electrospray ionization source (Agilent Technologies Inc., Santa Clara, CA, USA) and an Agilent 1290 series LC system (Agilent). The MS was operated in positive mode using the following conditions: gas temperature 250 °C, gas flow 12 L min^{-1}, nebulizer pressure 20 psi, sheath gas temperature

400 °C, sheath gas flow 12 L min^{-1}, capillary 3500 V, nozzle 0 V, high-pressure radio frequency 150 V, and low-pressure radio frequency 60. The analyses were performed using dynamic multiple reaction monitoring, monitoring a primary ion transition and a qualifier. The autosampler was maintained at 4 °C. Chromatographic separation of carotenoids was achieved on an Agilent Zorbax Bonus RP column (2.1 mm × 50 mm, 1.8 µm) maintained at 30 °C (±1 °C) at a flow rate of 600 µL min^{-1} with a 13 min run time. Injection volume was 3 µL. The mobile phase was composed of mobile phase A (0.1% formic acid in water) and mobile phase B (0.1% formic acid in methanol), with LC gradient: 5 min gradient from 78 to 85% mobile phase B, followed by 1.5 min from 85 to 100% B, then a 4.5 min hold at 100% B and 2 min re-equilibration to 78% B.

Chromatographic separation of pteridines was achieved on a Waters Acquity UPLC BEH C8 column (2.1 mm × 100 mm, 1.7 µm) maintained at 22 °C (±1 °C) at a flow rate of 300 µL min^{-1} with an 8 min run time. The injection volume was 1 µL. The mobile phase was composed of mobile phase A (0.1% formic acid in water) and mobile phase B (0.1% formic acid in methanol), with LC gradient: 3 min isocratic elution at 1% mobile phase B, followed by 0.5 min gradient to 90% B, then a 2.5 min hold at 90% B and 2 min re-equilibration to 1% B.

Data analysis was conducted in Agilent MassHunter Quantitative Analysis (version B.07.00), and all peak assignments were matched against commercial or purified (drosopterin) standards and confirmed with a qualifier ion (see retention times and other monitoring parameters in Supporting Information, Table S1). We used the following carotenoid standards: canthaxanthin, β-carotene, β-cryptoxanthin (Sigma-Aldrich, St Louis, MO, USA), 3′-dehydrolutein, astaxanthin, lutein and zeaxanthin (Sapphire Biosciences). Pteridine standards used: xanthopterin (Chem-Supply, Port Adelaide, SA, Australia), isoxanthopterin, pterine, pterine-6-carboxylic acid, 6-biopterin (Sigma-Aldrich), sepiapterin (Sapphire Biosciences) and drosopterin extracted and purified from fruit flies, *Drosophila melanogaster* (Wilson & Jacobson, 1977). Concentrations of metabolites were quantified from the relative response from individual calibration curves (internal calibration using the respective ISTD, total range across all eight calibration points: 0.00064 to 50 µM; minimum four-point calibration subset used: $R^2 > 0.95$). Owing to the peak overlap, concentrations of the isomer pair lutein/zeaxanthin were estimated from a zeaxanthin calibration curve.

Male morphology

We captured 61 males by noosing or by hand, and the majority of lizards were caught during night-time sampling when lizards were asleep. Only males with SVL > 215 mm were included here, because this is the minimal size at which they are sexually mature (Thompson, 1993). We measured SVL from the tip of the snout to the posterior tip of the vent and tail length from the end of the vent to the tail tip. Both measurements were made using a plastic ruler to the nearest 1 mm. We measured four head dimensions: head width, the minimal distance separating the left and right tympanum; head height, from the base of the crest to the underside of the jaw with the mouth held closed; maximal head width, the widest distance between cheek muscles with the head of the lizard flat on a table; and head length, from the tip of the snout to the beginning of the tympanum on the right side of the head. All measurements were taken with digital callipers to the nearest 0.01 mm. All lizards were weighed on a digital scale to the nearest 1 g. Summary statistics for all morphological variables are given in the Supporting Information (Table S2). We also counted the number of ticks on every lizard. Before release, all lizards were individually marked using PIT tags. Lizards were typically released on the day after capture.

We calculated body condition using the residuals of a regression of body mass on SVL ($F_{1,59} = 46.27$, $P < 0.0001$, $r^2 = 0.43$). We conducted a principal component analysis (PCA) on the four head dimensions because they were highly correlated with each other. We used the score on the first principal component (which explained 72% of the variance) for further analyses involving head dimensions (loadings: head length, 0.43; head height, 0.52; widest head width, 0.51; and head width, 0.53); this measure will be referred to as head size. The PCA was run on a correlation matrix using the *ade4* package (Dray & Dufour, 2007). Although most studies standardize head dimensions by taking residuals of head size on SVL, we opted to use the original values for two reasons. First, the resource-holding potential of water dragon males increases with isometric growth instead of allometric growth, i.e. larger males have larger (non-standardized) heads and are more likely to be territorial (Baird *et al.*, 2012). Second, non-corrected measurements of head size and bite force are significantly correlated in males from this population (D.C., unpubl. data).

Male coloration and visual modelling

Spectral reflectance curves were obtained using an Ocean Optics spectrometer (USB2000; Ocean Optics Inc., Dunedin, FL, USA) with a PX2 light source. All measurements were taken of a 6 mm^2 region of skin at a constant distance of 5 mm, at an angle of 45° using an RPH-1 probe holder in the laboratory. We took readings from six different body regions: forelimb, throat (centre of the throat), throat-base, chest, abdomen and thighs. All readings were relative to a dark and

99% white (WS-1) standard. We measured forelimb and thigh coloration because we found these regions to be highly variable in the extent and intensity of red coloration displayed by males. The measurements were taken from the ventral surface because this is where any red colour would be expressed. Likewise, we focused on two separate regions of the throat owing to variation in the range of red coloration between males; some males had red coloration spreading up through the base of the throat, whereas others did not. In the case of the chest, we took measurements in the area below the forelimbs and above the mid-point between the fore- and hindlimbs. In all but these two locations (throat and throat-base), two readings were taken on opposite sides. The average of the two readings (where applicable) over the range 300–700 nm was used in the analysis (Supporting Information, Fig. S2A, B).

Given that visual systems encode signals as contrasts between adjacent colours, we analysed male colour variation as changes in the contrast (relative to the lizard visual system) between male body region colours and the habitat background. We sampled habitat background in relationship to its availability. The dominant background was the bark of trees, which is also a major component of the ground cover, and the leaves of trees and bushes. We therefore estimated background reflectance of the habitat from 33 reflectance curves of bark, 60 of leaves, six of rocks and six of sand samples measured in the environment of the lizards. Taking an average viewing background is a common approach in signalling studies where animals have been observed from a wide range of locations in the environment (e.g. Whiting *et al.*, 2015). The average reflectance was calculated (Supporting Information, Fig. S2C) and used to calculate the contrasts against male body region colours. However, we also ran the analyses while considering a bark background only or a leaf background only to see whether different backgrounds influenced results, but they did not. Hence, we show the results with the average background reflectance.

Following previous work on lizard coloration (Teasdale *et al.*, 2013; Merkling *et al.*, 2015), we used the Vorobyev–Osorio model to estimate these contrasts (Vorobyev & Osorio, 1998; Osorio & Vorobyev, 2008). This model estimates the discriminability of two colours in units of just noticeable differences (JNDs) and assumes that visual discrimination is limited by photoreceptor noise. It produces a measure of the chromatic (colour) and achromatic (luminance) contrasts based on the four single cones used for colour perception and the double cone used for luminance perception, respectively (Vorobyev & Osorio, 1998). We used model calculations detailed by Siddiqi *et al.* (2004) and calculated the contrasts using the same methods as Merkling *et al.* (2015). More details are given in Methods part of the Supporting Information. Briefly, we first calculated the relative stimulation of photoreceptors for each body region relative to the background, while taking into account the visual system of the dragons (using data from a closely related species; Barbour *et al.*, 2002). Then we calculated chromatic and achromatic contrasts for each body region by accounting for photoreceptor noise and ratio. Visual modelling analyses thus resulted in 12 colour variables. Contrasts greater than one mean that the focal colour can be discriminated from the background by the animals. All analyses were done using the R package *pavo* (Maia *et al.*, 2013).

Given that some colour variables were strongly correlated, we reduced the number of variables by using PCA scores. Abdomen, forelimb, chest and thigh chromatic contrasts were correlated (all $r > 0.4$); the first principal component of a PCA on these four variables explained 66% of the variance (loadings: abdomen, 0.47; forelimb, 0.50; chest, 0.52; and thigh, 0.51) and will be referred to as PCA chromatic. The achromatic contrasts of the same variables were also correlated (all $r > 0.68$): the first principal component of a PCA explained 82% of the variance (loadings: abdomen, 0.51; forelimb, 0.49; chest, 0.50; and thigh, 0.50) and will be referred to as PCA achromatic. We kept the contrasts for throat and throat-base apart because they were not strongly correlated with the other variables or among themselves ($r = 0.27$ among themselves, and 0.2 less than the other variables were correlated among themselves). We thus ended up with six colour variables.

Statistical analyses

To clarify the signalling role of red coloration in male water dragons, we ran linear regressions with SVL, mass, body condition or head size as response variables and the six colour variables as predictor variables. The SVL, mass, body condition and head size were significantly correlated, and some correlations were fairly strong (maximal $r = 0.75$ between mass and body condition). We still chose to consider all of them as response variables to see if the same colour variables were correlated with them, but we took this collinearity into account in our interpretation of the results.

Parasite load (number of ticks) was analysed using a negative binomial generalized linear model (GLM) in the *MASS* package (Venables & Ripley, 2002), because the data were overdispersed (overdispersion parameter = 15) when using a Poisson distribution.

Model selection was based on the information-theoretic approach, which allows the consideration of non-nested models and to quantify their relative goodness of fit to the data based on the corrected Akaike information criterion (AICc) (Burnham & Anderson, 2002; Burnham *et al.*, 2011). As our sample size was fairly small and to avoid model overfitting, we considered all possible models with one or two predictors and no interactions, which led to a final list of 22 models (including one with intercept only). For each model, we calculated its AICc and its AICc difference (ΔAICc) with the lowest AICc model (corresponding to the best model). The ΔAICc provides a measure of information lost between the best model and a given model. Following Burnham *et al.* (2011), we selected the models with ΔAICc < 4 instead of the more classical < 2 threshold. This almost always led to a cumulative weight of 0.95, therefore making our inferences more robust. We also calculated a weight (ωAICc) for each model, which represents the probability that a given model is the best approximating model among the subset of models (Symonds & Moussalli, 2011). These parameters were calculated using the *AICcmodavg* package (Mazerolle, 2013) for the linear regressions and the *MuMIn* package (Barton, 2016) for the negative binomial GLMs. As this approach resulted in multiple models that were equally likely, we computed model-averaged parameter estimates, standard errors, confidence intervals and the relative importance value (i.e. sum of ωAICc of the models containing the variable) without shrinking the parameters (Burnham & Anderson, 2002). Briefly, we averaged the parameter estimates of each variable for all the models in which they appeared, while taking into account their relative weight. This allowed us to take model selection uncertainty into account. As a further measure of goodness of fit, adjusted R^2 values are given in the Supporting Information, Tables S5–S8 for all models in the subset. All analyses were run using R 3.2.2 (R Core Team, 2016).

RESULTS

Source of colour: carotenoids vs. pteridines

Mass spectrometry revealed that both carotenoids and pteridines were present in the red ventral and brown dorsal skin of water dragons, but pteridines were the predominant contributors in the red skin (Fig. 1; and Supporting Information, Table S3 for a complete list of the carotenoids and pteridines identified and the colour they produce). Concentrations of carotenoids were similar in red and brown skin (Fig. 1A). However, concentrations of pteridines were much higher than those of carotenoids, and the concentrations of the coloured pteridines drosopterin and isoxanthopterin were substantially higher in red than brown skin (Fig. 1B). Most notably, the concentration of red drosopterin was > 700 times higher in red than brown skin and many orders of magnitude higher than other coloured pteridines. Taken together, these data indicate that the red ventral coloration in water dragons is predominantly generated by drosopterin.

Male morphology and colour signals

All aspects of the red ventral coloration of males could be discriminated against the background, because all contrasts were greater than one and had a confidence interval that did not overlap with this value (Supporting Information, Table S4).

No colour variables predicted variation in SVL (Supporting Information, Table S5). Although the best model contained the achromatic contrast of throat-base, the null model (intercept only) was equally likely. However, the null model was not in the four top models explaining mass, whereas PCA chromatic was present in all of them (Supporting Information, Table S6). Heavier males were less conspicuous than smaller ones (Table 1; Fig. 2). Throat achromatic contrast had a positive effect on mass (Table 1), but given that the effect was small (i.e. small estimate and confidence interval close to zero) and that it appeared in only one of the top models this suggests that it was not a strong predictor of mass.

For body condition, males in better condition were less conspicuous (PCA chromatic) than those in lower condition, similar to the effect found for mass (Table 2; Fig. 3). The PCA chromatic was present in all of the top models (Supporting Information, Table S7), whereas the null model was in none of them (Supporting Information, Table S7). The PCA achromatic and throat chromatic contrasts both had a positive effect on body condition (Table 2), but given that they appeared in only one of the top models this suggests that they were not strong predictors of body condition.

For head size, males with larger heads were less conspicuous, although the effect was rather small (i.e. small estimate and confidence interval slightly overlaps zero; Table 3). The best model contained PCA chromatic, but 11 models had a ΔAICc < 4, and each colour variable appeared at least once in these models (Supporting Information, Table S8).

Parasite loads and colour signals

Twenty-three (38%) of 61 males had one or more ticks, and the individuals with the more conspicuous red coloration (PCA chromatic) had a higher parasite load (Table 4; Fig. 4). The PCA chromatic was in all the top models (Supporting Information, Table S9), whereas although base-throat chromatic contrast also had a confidence interval not overlapping zero (Table 4), it appeared only in the top model with PCA chromatic (Supporting Information, Table S9) and therefore had a weak effect. Two individuals had very high tick counts, so we reanalysed the data after excluding them to check whether the effect of PCA chromatic remained. More conspicuous individuals still had a higher parasite load, but the effect was weaker [estimate ± SE: 0.44 ± 0.18 (0.08; 0.80)].

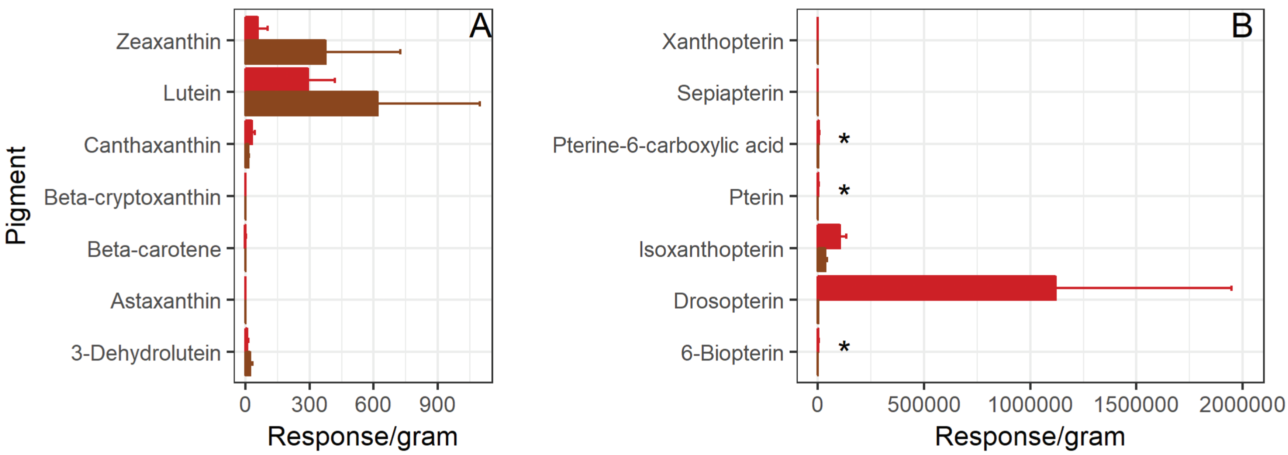

Figure 1. Distribution of carotenoids (A) and pteridines (B) in the brown and red skin of water dragons ($N = 3$). Note the scale difference between carotenoids and pteridines. Colourless pteridines are indicated by an asterisk. More information about the colour produced by each pigment is given in the Supporting Information (Table S3).

Table 1. Model-averaged estimates for all colour variables in the subset of models explaining mass in water dragons

Parameter	Estimate	SE	Lower CL	Upper CL	Importance
Intercept	640.33	71.81	499.59	781.07	–
PCA chromatic*	−29.55	8.24	−45.70	−13.40	1
PCA achromatic	15.49	8.38	−0.94	31.92	0.29
Throat achromatic contrast*	7.07	3.41	0.39	13.749	0.45
Base-throat achromatic contrast	3.53	3.28	−2.89	9.95	0.09

Estimate, model-averaged estimate of the variable; importance, relative importance of the variable; lower CL and upper CL, lower and upper limit of the 95% confidence interval; SE, unconditional standard error of the parameter.
*Estimates different from zero.

DISCUSSION

The mechanisms and function of coloration are still poorly understood in reptiles. Species with yellow to red colours are particularly interesting because two very different classes of pigments (carotenoids and pteridines) can be responsible for their production. In this study, we investigated the mechanisms and function of the red coloration in male water dragons, a sexually dimorphic lizard where males are known to engage in intense fights (Baird *et al.*, 2014). Their red hue was mostly attributable to the presence of red-coloured pteridines (primarily drosopterin, but also xanthopterin and sepiapterin), whereas carotenoids were in much lower concentrations. Animals can synthesize pteridines *de novo*, and evidence of any associated costs is tentative so far. Hence, the signalling role of pteridine-based ornaments is still unclear. Here, we showed that some aspects of male red coloration were related to key morphological variables that are known to be important in male contest competition for many lizard species (e.g. Stuart-Fox & Ord, 2004). We predicted that if red coloration was a signal of body condition or fighting ability, it would be positively correlated with morphology. In contrast, we found that the chromatic contrast between red body regions (abdomen, forelimbs, chest and thighs) and the habitat background were negatively correlated with mass, body condition and head size. It is not surprising to find similar results for different morphological response variables, because they were positively correlated (especially mass and body condition). Surprisingly, chromatic contrast did not explain variation in SVL, although it was also correlated with the other morphological variables. None of the other colour variables was correlated with any morphological variables, thus suggesting that the coloration of the throat and throat-base might not act as a signal in the traditional sense or signals something that we did not measure. However, heavier males in better condition seemed to have more melanin within the red colour patch (black spots interspersed with darker red; see Supporting Information, Fig. S1), suggesting that the inverse relationship could reflect a melanin-based signal. Finally, we investigated whether the most heavily parasitized individuals were the less intensely coloured, as would be expected if there was a trade-off between immunity and coloration. Contrary to this prediction, we found that the individuals with the most conspicuous red body regions (abdomen, forelimbs, chest and thighs) against the habitat background had a higher number of ticks, although the effect was weak.

Pteridines are the main pigments responsible for the red coloration in other reptiles, including the lizards *Anolis sagrei* (Steffen & McGraw, 2009), *Anolis pulchellus* (Ortiz *et al.*, 1962; Ortiz & Williams-Ashman, 1963) and *Sceloporus virgatus* (Weiss *et al.*, 2011, 2012). Some studies have suggested that pteridines may be costly to produce, and that they can act as cofactors for enzymatic reactions, but whether organisms face a trade-off between these proposed alternative functions of pteridines and pigment production is unknown (Ortiz & Williams-Ashman, 1963; Stackhouse, 1966; Grether *et al.*, 2001). Pteridine-based colour patches signal the antioxidant levels in egg yolks of female striped plateau lizards (*S. virgatus*), presumably to avoid resource trade-offs and allow for use of carotenoids in an antioxidant function (Weiss *et al.*, 2011). Could redder coloration attributable to more pteridines likewise signal higher antioxidant capacity in water dragons? Given the inverse relationship between conspicuousness and body mass, this is not necessarily a straightforward relationship and would need to be addressed by quantifying the covariation of pteridines in the blood and skin, which was beyond the scope of the present study. Conversely, signals could be cheap to produce but still impose maintenance costs and thereby signal quality. For example, only high-quality individuals might be able to bear the cost of increased conspicuousness, or expression might be a function of stress levels. Until additional data become available on the potential costs of pteridine production and maintenance, our understanding of the honesty and function of pteridine-based signals in reptiles will remain limited.

Given that mass, body condition and head size are important predictors of contest outcome and fighting ability in numerous lizard species (Whiting *et al.*, 2003; Baird, 2013), coloration of the abdomen, forelimbs, chest and thighs has the potential to convey information about individual fighting or competitive ability and might act as a status signal (Whiting *et al.*, 2003) in water dragons. We thus predicted that larger males, with greater fighting ability, would have a more conspicuous red coloration. However, heavier individuals, in better condition and with larger heads, had less intense red. This means that these individuals were less conspicuous and better at matching the colour of the background, which is opposite to our prediction. One explanation for this counter-intuitive result is that the chests of heavier males might be darker because of the accumulation of melanin (see Supporting Information, Fig. S1), the widespread pigment responsible for black coloration across taxa. The deposition of melanin on to a colour patch as males

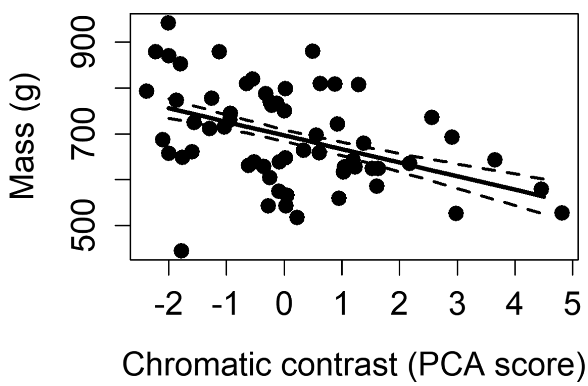

Figure 2. Mass in relationship to chromatic contrasts of colour patches against the habitat background [r = −0.41 (−0.60; −0.17)]. Chromatic contrasts are scores on the first principal component of a principal component analysis (PCA) on the chromatic contrasts of abdomen, forelimbs, chest and thighs. Continuous line represents the model-averaged predictions and dashed lines their standard errors.

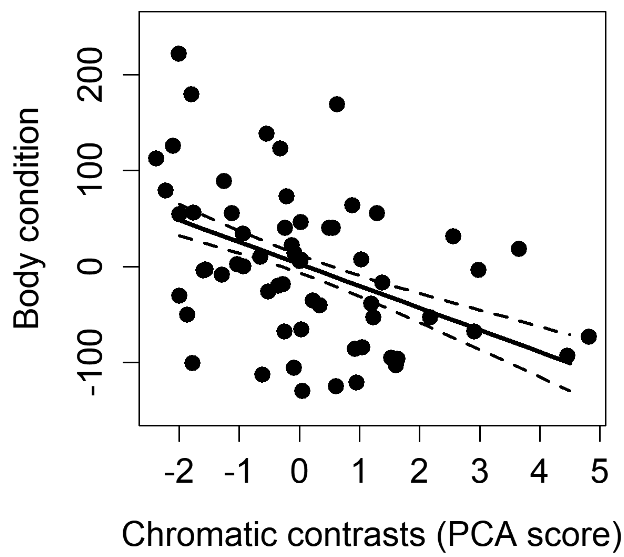

Figure 3. Body condition in relationship to chromatic contrasts of colour patches against the habitat background [r = −0.41 (−0.60; −0.17)]. Chromatic contrasts are scores on the first principal component of a principal component analysis (PCA) on the chromatic contrasts of abdomen, forelimbs, chest and thighs. Continuous line represents the model-averaged predictions and dashed lines their standard errors.

Table 2. Model-averaged estimates for all colour variables in the subset of models explaining body condition in water dragons

Parameter	Estimate	SE	Lower CL	Upper CL	Importance
Intercept	−29.06	49.57	−126.22	68.09	–
PCA chromatic*	−22.90	6.36	−35.36	−10.44	1
PCA achromatic*	13.60	6.20	1.44	25.76	0.41
Throat achromatic contrast*	5.28	2.55	0.28	10.28	0.32
Base-throat achromatic contrast	2.83	2.45	−1.90	7.69	0.07
Throat chromatic contrast	18.78	16.69	−13.94	51.50	0.07

Estimate, model-averaged estimate of the variable; importance, relative importance of the variable; lower CL and upper CL, lower and upper limit of the 95% confidence interval; SE, unconditional standard error of the parameter.
*Estimates different from zero.

Table 3. Model-averaged estimates for all colour variables in the subset of models explaining head size in water dragons

Parameter	Estimate	SE	Lower CL	Upper CL	Importance
Intercept	0.51	1.18	−1.81	2.82	–
Throat-base achromatic contrast	−0.09	0.07	−0.23	0.03	0.23
Throat-base chromatic contrast	−0.59	0.37	−1.30	0.13	0.34
PCA achromatic	−0.24	0.15	−0.53	0.04	0.31
PCA chromatic*	−0.34	0.14	−0.61	−0.07	0.69
Throat achromatic contrast	0.05	0.06	−0.07	0.17	0.11
Throat chromatic contrast	0.07	0.37	−0.65	0.79	0.06

Estimate, model-averaged estimate of the variable; importance, relative importance of the variable; lower CL and upper CL, lower and upper limit of the 95% confidence interval; SE, unconditional standard error of the parameter.*Estimates different from zero.

Table 4. Model-averaged estimates for all colour variables in the subset of models explaining ectoparasite load (i.e. number of ticks) in water dragons

Parameter	Estimate	SE	Lower CL	Upper CL	Importance
Intercept	1.53	1.67	−1.75	4.81	–
Base-throat chromatic contrast*	−1.21	0.52	−2.22	−0.20	0.59
PCA chromatic*	0.76	0.23	0.32	1.21	1
Base-throat achromatic contrast	0.16	0.16	−0.15	0.46	0.1
Throat achromatic contrast	0.13	0.17	−0.20	0.47	0.09

Estimate, model-averaged estimate of the variable; importance, relative importance of the variable; lower CL and upper CL, lower and upper limit of the 95% confidence interval; SE, unconditional standard error of the parameter.*Estimates different from zero.

age has also been documented in the western fence lizard, *Sceloporus occidentalis* (Ressel & Schall, 1989). And, in Rawson lizards, *Liolaemus xanthoviridis*, larger, and therefore older, males were more melanistic (Escudero *et al.*, 2016). Moreover, studies on species of the genus *Sceloporus* have revealed that melanism is positively related to testosterone (Quinn & Hews, 2003; Cox *et al.*, 2005, 2008) and a study on the common lizard, *Zootoca vivipara*, suggested that more melanistic individuals could mount a higher immune response (Vroonen *et al.*, 2003). The fact that darker individuals are heavier and in better condition and have larger heads suggests that in water dragons, melanin might be the primary pigment signalling some aspect of male quality, as opposed to pteridines. If water dragons accumulate melanin at the expense of pteridines as they age, melanin could act as an honest signal of age and, potentially, quality. However, we did not measure how melanin and pteridine concentrations change with age, size and condition; therefore, this is merely speculation. Future studies of alternative reproductive tactics should test whether levels of melanin predict social strategy and fitness.

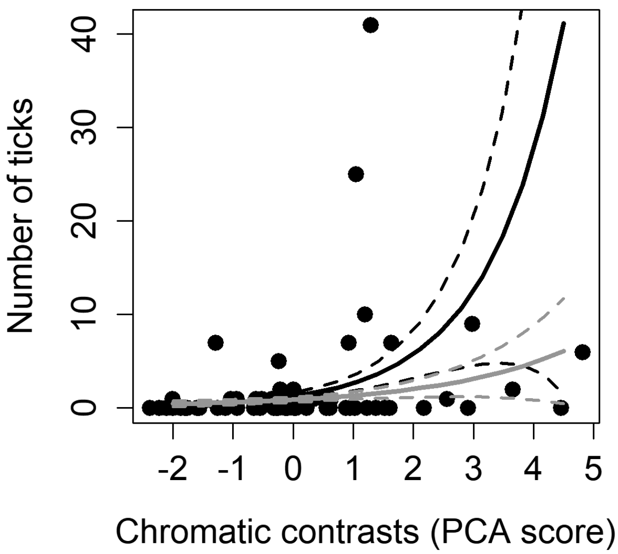

Figure 4. Number of ticks in relationship to chromatic contrasts of colour patches against the habitat background. Chromatic contrasts are scores on the first principal component of a principal component analysis (PCA) on the chromatic contrasts of abdomen, forelimbs, chest and thighs. The continuous black line represents the model-averaged predictions on the full dataset and dashed lines their standard errors. The grey lines are predictions and standard errors after excluding the two individuals with high tick count.

Interestingly, male water dragons with a higher parasite load had a more conspicuous red coloration. This is, for example, contrary to what has been found in other lizards, such as *L. schreiberi* (Megía-Palma *et al.*, 2017). This could represent a trade-off between resources allocated to coloration and immunity. However, the inverse relationship between body size and chromatic contrast against the background makes this unlikely. Given our interpretation that redder males are younger, the higher number of ticks in these individuals could reflect a less developed immune system and/or lower-quality individuals that have not yet survived to an age and size at which they are darker (selective disappearance; Curio, 1983).

Contests between males can be intense (Baird *et al.*, 2013, 2014), resulting in open wounds that form lasting scars. These costs have the potential to be ameliorated if a signal of the competitive ability of a male is available to a rival. The correlation between male red coloration and morphological variables further supports the potential signalling role of ventral coloration. Furthermore, the ventral location of coloration in water dragons also suggests an evolutionary balance or trade-off between sexual signalling and survival via camouflage (Stuart-Fox *et al.*, 2004). Additional studies using staged contests will help to disentangle the relative roles of colour and morphology as signals of male fighting ability and quality.

In summary, we found that although the coloration in water dragons is the result of both pteridine and carotenoid pigments, the distinctive red is driven primarily by drosopterin. However, heavier males in better condition and with larger heads were less conspicuously red, which might be attributable to the accumulation of melanin. This result is particularly interesting because it is the opposite to what we predict in sexual selection theory, i.e. that more exaggerated (conspicuous) traits signal male quality. In the case of the water dragons, head size and weight, which are classic indicators of quality in lizards, were inversely related to conspicuousness. Nevertheless, colour does signal male quality, and our results highlight a potential role for melanin as a sexual signal in lizards, including those that also contain conspicuous pigments such as pteridines.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Table S1. Dynamic multiple reaction monitoring parameters.
Table S2. Summary statistics of the six morphological variables measured on male water dragon lizards.
Table S3. Pigments identified in the red ventral and/or brown dorsal skin of male water dragons according to their class (carotenoid or pteridine) and the colour they produce.

Table S4. Summary statistics of the 12 colour contrasts measured on male water dragon lizards.
Table S5. The subset of 17 models with a ΔAICc < 4 relative to the best model among the 22 models considered to explain SVL in water dragons.
Table S6. The subset of four models with a ΔAICc < 4 relative to the best model among the 22 models considered to explain body mass in water dragons.
Table S7. The subset of five models with a ΔAICc < 4 relative to the best model among the 22 models considered to explain body condition in water dragons.
Table S8. The subset of 11 models with a ΔAICc < 4 relative to the best model among the 22 models considered to explain head size in water dragons.
Table S9. The subset of four models with a ΔAICc < 4 relative to the best model among the 22 models considered to explain the probability of being infected by ticks in water dragons.
Figure S1. Adult male water dragons (*Intellagama lesueurii*) from Lane Cove National Park. The male on the left is an older male with a darker chest than the younger male on the right.
Figure S2. A, abdomen, forelimb, chest and thigh average (± SE) reflectance. B, throat average (± SE) reflectance. C, average background reflectance. D, lizard spectral sensitivities. Abbreviations: D, double cone; LWS, long-wavelength-sensitive single cone; MWS, medium-wavelength-sensitive single cone; SWS, short-wavelength-sensitive single cone; UVS, ultraviolet-sensitive single cone; Barbour *et al.* (2002).

SHARED DATA

All data and scripts used in this study are available at the Open Science Framework: https://osf.io/n6eyt/<https://protect-au.mimecast.com/s/M7qLCXLW6Di895yc6V1r8?domain=osf.io>

Supplementary Material

Seeing red: pteridine-based colour and male quality in a dragon lizard

Thomas Merkling, Dani Chandrasoma, Katrina J. Rankin, and Martin J. Whiting

Methods

Male Colouration and visual modelling

First, we calculated the relative stimulation of photoreceptors (receptor quantum catches) in a lizard visual system for all male body region colours using the equation detailed in Vorobyev & Osorio (1998), following Teasdale *et al.* (2013). This analysis resulted in five variables (quantum catches) for each body region, one for each of the four single cones used for colour perception and the double cone which is thought to be used for luminance perception (Osorio & Vorobyev, 2005). Spectral sensitivities are not known for water dragons, but visual systems are phylogenetically conserved within lizard families (Olsson *et al.*, 2013), so we used spectral sensitivity data of a closely related agamid, *Ctenophorus ornatus*, where three single visual pigment types (short, medium, and long wavelength-sensitive) have been identified (Figure S1d; Barbour *et al.*, 2002). The authors, however, failed to identify an ultraviolet-sensitive (UVS) photoreceptor that has been found in all other related lizards (Loew *et al.*, 2002; Bowmaker *et al.*, 2005). Evidence of the UVS photoreceptor has recently been found in another closely related agamid, *Ctenophorus decresii*, as four opsin genes were expressed (Yewers *et al.*, 2015). Therefore, we considered an additional fourth UVS photoreceptor ($\lambda max = 360$ nm; Figure S1d, following Chan *et al.*, 2009). We also used irradiance spectra representing a shady forest habitat, as this is more representative of water dragon habitat. However, choosing an irradiance spectrum representing full sunlight did not influence the results. We applied the von Kries transformation to the cone catches to normalise them to the background, thus accounting for receptor adaptation to the light environment (Vorobyev & Osorio, 1998).

Photoreceptor noise and photoreceptor ratio were used following Teasdale *et al.* (2013). For chromatic contrasts, we assumed a Weber fraction (ω_i) for the LWS photoreceptor = 0.1 and we used a ratio of 1 : 1: 3.5 : 6 for the four single receptor classes (Barbour *et al.*, 2002), hence the signal-to-noise ratio was 0.072. We then derived ω_i for the remaining photoreceptor classes (Stuart-Fox *et al.*, 2003; Siddiqi *et al.*, 2004). For achromatic contrasts, since double cones are usually more abundant than any of the single cones including LWS, we assumed a Weber fraction (ω_i) for the LWS photoreceptor = 0.05 and thus had to set the relative number of LWS cone to 4 and the noise-to-signal ratio to 0.1. Analyses were run with the *pavo* R package (Maia *et al.*, 2013) and examples of how contrasts were calculated are given in Merkling *et al.* (2015).

Supplementary Tables

Table S1. Dynamic multiple reaction monitoring parameters.

		Main ion			Qualifier		
Carotenoids	**RT**	**Q1**	**Q3**	**CE**	**Q1**	**Q3**	**CE**
β-apo-8'-carotenal (ISTD)	4.8	416.3	91.1	60	416.3	210	24
Astaxanthin	1.8	597.4	147.1	20	597.4	118.9	60
3'-Dehydrolutein	3.3	566.4	159.1	36	566.4	105.0	56
Lutein/Zeaxanthin	4.3	568.4	476.3	12	568.4	119.0	56
Canthaxanthin	4.8	565.4	203.0	20	565.4	132.9	36
β-Carotene	8.4	536.4	444.3	12	536.4	104.9	60
β-Cryptoxanthin	7.2	552.4	105.0	60	552.4	119.0	56
Pteridines	**RT**	**Q1**	**Q3**	**CE**	**Q1**	**Q3**	**CE**
Biopterin-d3 (ISTD)	2.9	241.1	223.0	12	241.1	179.1	20
Xanthopterin	1.6	180.1	135.1	20	180.1	163.1	16
6-Biopterin	2.9	238.1	219.7	12	238.1	178.0	20
Pterin	2.0	164.1	119.1	20	164.1	92.1	32
Isoxanthopterin	2.0	180.1	135.0	20	180.1	110.1	28
Pterine-6-Carboxylic Acid	1.2	208.1	190.0	12	208.1	162.0	16
Sepiapterin	4.9	238.1	192.0	12	238.1	165.2	20
Drosopterin	4.8	369.1	218.0	28	369.1	229.9	28

Q1, precursor ion (m/z); Q3, product ion (m/z); CE, collision energy (eV); RT, retention time (min); ISTD, internal standard

Table S2. Summary statistics of the six morphological variables measured on male water dragon lizards.

Variable	Mean	Standard error	Minimum	Maximum
Mass (g)	692.9	13.72	444.8	942.1
SVL (mm)	236.6	1.15	216	255
Head length (mm)	54.27	0.51	36.9	61.13
Head width (mm)	41.56	0.42	28.95	47.6
Head width widest (mm)	69.21	0.91	47.79	85.39
Head height (mm)	39.40	0.55	25.36	49.62

Table S3. Pigments identified in the red ventral and/or brown dorsal skin of male water dragons according to their class (carotenoid or pteridine) and the colour they produce.

Class	Pigment	Colour produced
Carotenoid	3'-Dehydrolutein (both)	Yellow
	Astaxanthin (red)	Red
	β-Carotene (both)	Orange-Red
	β-Cryptoxanthin (both)	Yellow
	Canthaxanthin (both)	Orange-Red
	Lutein (both)	Yellow
	Zeaxanthin (both)	Orange-Red
Pteridine	6-Biopterin (both)	Colourless/UV
	Drosopterin (both)	Red
	Isoxanthopterin (both)	Purple
	Pterin (both)	Colourless/UV
	Pterine-6-Carboxylic Acid (both)	Colourless/UV
	Sepiapterin (red)	Yellow
	Xanthopterin (both)	Yellow

Table S4. Summary statistics of the twelve colour contrasts measured on male water dragon lizards. 'SE' denotes the standard error of the variable; 'Lower CL' and 'Upper CL' are the lower and upper limit of the 95% confidence interval.

Colour contrast	Mean	SE	Lower CL	Upper CL
Abdomen chromatic contrast	1.86	0.04	1.78	1.94
Forelimb chromatic contrast	2.35	0.10	2.15	2.55
Chest chromatic contrast	2.42	0.09	2.25	2.59
Thigh chromatic contrast	2.23	0.09	2.06	2.40
Throat chromatic contrast	1.62	0.07	1.47	1.76
Base-throat chromatic contrast	2.08	0.09	1.89	2.26
Abdomen achromatic contrast	4.02	0.20	3.63	4.41
Forelimb achromatic contrast	3.02	0.22	2.60	3.44
Chest achromatic contrast	3.18	0.23	2.73	3.62
Thigh achromatic contrast	4.44	0.23	4.00	4.89
Throat achromatic contrast	8.46	0.23	8.00	8.91
Base-throat achromatic contrast	3.93	0.25	3.43	4.43

Table S5. The subset of 17 models with a ΔAICc < 4 relative to the best model among the 22 models considered to explain SVL in water dragons. 'K' denotes the number of parameters; 'AICc' is the AIC corrected for finite sample size; 'ΔAICc' is the difference between AICc of a given model to that of the best model; 'ωAICc' is the probability of each model given the data and the model set; 'logLik' is the log-likelihood; 'Cum. ωAICc' is the cumulative probability; 'R^2' is the adjusted R-squared.

Model	K	AICc	ΔAICc	ωAICc	logLik	Cum. ωAICc	R^2
Base-throat achromatic contrast	3	442.92	0	0.16	-218.25	0.16	0.03
Null	2	443.73	0.8	0.11	-219.76	0.26	0
Base-throat chromatic contrast + Throat achromatic contrast	4	444.39	1.47	0.08	-217.84	0.34	0.03
PCA Chromatic	3	444.46	1.54	0.07	-219.02	0.41	0.02
Base-throat chromatic contrast + PCA Chromatic	4	444.86	1.94	0.06	-218.07	0.47	0.02
Base-throat achromatic contrast + Base-throat chromatic contrast	4	444.92	2	0.06	-218.1	0.53	0.02
Base-throat chromatic contrast + Throat chromatic contrast	4	445.15	2.23	0.05	-218.22	0.58	0.02
Base-throat chromatic contrast + PCA Achromatic	4	445.17	2.25	0.05	-218.23	0.64	0
Throat chromatic contrast	3	445.32	2.4	0.05	-219.45	0.68	0
Throat achromatic contrast	3	445.57	2.65	0.04	-219.58	0.73	0
PCA Achromatic	3	445.9	2.98	0.04	-219.74	0.76	0
Base-throat achromatic contrast	3	445.94	3.02	0.03	-219.76	0.8	0
Throat achromatic contrast + PCA Chromatic	4	446.22	3.3	0.03	-218.76	0.83	0

Throat chromatic contrast + PCA Chromatic	4	446.28	3.35	0.03	-218.78	0.86	0
Throat achromatic contrast Throat chromatic contrast	4	446.55	3.63	0.03	-218.92	0.88	0
PCA Achromatic + PCA Chromatic	4	446.65	3.73	0.02	-218.97	0.91	0
Base-throat achromatic contrast + PCA Chromatic	4	446.67	3.75	0.02	-218.98	0.93	0

Table S6. The subset of 4 models with a ΔAICc < 4 relative to the best model among the 22 models considered to explain body mass in water dragons. 'K' denotes the number of parameters; 'AICc' is the AIC corrected for finite sample size; 'ΔAICc' is the difference between AICc of a given model to that of the best model; 'ωAICc' is the probability of each model given the data and the model set; 'logLik' is the log-likelihood; 'Cum. ωAICc' is the cumulative probability; 'R^2' is the adjusted R-squared.

Model	**K**	**AICc**	**ΔAICc**	**ωAICc**	**logLik**	**Cum. ωAICc**	**R^2**
PCA Chromatic + Throat achromatic contrast	4	735.59	0	0.40	-363.44	0.40	0.20
PCA Chromatic + PCA Achromatic	4	736.47	0.88	0.26	-363.88	0.66	0.19
PCA Chromatic	3	737.66	2.08	0.14	-365.62	0.80	0.15
PCA Chromatic + Base-throat achromatic contrast	4	738.75	3.16	0.08	-365.02	0.88	0.15

Table S7. The subset of 5 models with a ΔAICc < 4 relative to the best model among the 22 models considered to explain body condition in water dragons. 'K' denotes the number of parameters; 'AICc' is the AIC corrected for finite sample size; 'ΔAICc' is the difference between AICc of a given model to that of the best model; 'ωAICc' is the probability of each model given the data and the model set; 'logLik' is the log-likelihood; 'Cum. ωAICc' is the cumulative probability; 'R^2' is the adjusted R-squared.

Model	K	AICc	ΔAICc	ωAICc	logLik	Cum. ωAICc	R^2
PCA Chromatic + PCA Achromatic	4	699.77	0	0.39	-345.53	0.39	0.20
PCA Chromatic + Throat achromatic contrast	4	700.27	0.50	0.31	-345.78	0.70	0.20
PCA Chromatic	3	702.33	2.56	0.11	-347.96	0.81	0.15
PCA Chromatic + Base-throat achromatic contrast	4	703.17	3.40	0.07	-347.23	0.88	0.16
PCA Chromatic + Throat chromatic contrast	4	703.31	3.54	0.07	-347.30	0.95	0.16

Table S8. The subset of 11 models with a ΔAICc < 4 relative to the best model among the 22 models considered to explain head size in water dragons. 'K' denotes the number of parameters; 'AICc' is the AIC corrected for finite sample size; 'ΔAICc' is the difference between AICc of a given model to that of the best model; 'ωAICc' is the probability of each model given the data and the model set; 'logLik' is the log-likelihood; 'Cum. ωAICc' is the cumulative probability; 'R^2' is the adjusted R-squared.

Model	K	AICc	ΔAICc	ωAICc	logLik	Cum. ωAICc	R^2
PCA Chromatic	3	235.55	0	0.17	-114.56	0.17	0.11
PCA Chromatic + PCA Achromatic	4	235.77	0.22	0.15	-113.53	0.32	0.13
Base-throat chromatic contrast + Base_throat achromatic contrast	4	236.12	0.57	0.13	-113.70	0.45	0.12
PCA Chromatic + Base-throat chromatic contrast	4	236.55	0.99	0.10	-113.92	0.55	0.11
PCA Chromatic + Base-throat achromatic contrast	4	236.86	1.31	0.09	-114.07	0.64	0.11
PCA Chromatic + Throat achromatic contrast	4	237.22	1.67	0.07	-114.25	0.71	0.10
PCA Achromatic + Base-throat chromatic contrast	4	237.65	2.10	0.06	-114.47	0.77	0.10
PCA Chromatic + Throat chromatic contrast	4	237.81	2.25	0.05	-114.54	0.82	0.10
PCA Achromatic	3	238.18	2.63	0.04	-115.88	0.86	0.07
PCA Achromatic + Throat achromatic contrast	4	239.16	3.60	0.03	-115.22	0.89	0.08
Base-throat chromatic contrast	3	239.34	3.78	0.02	-116.46	0.91	0.05

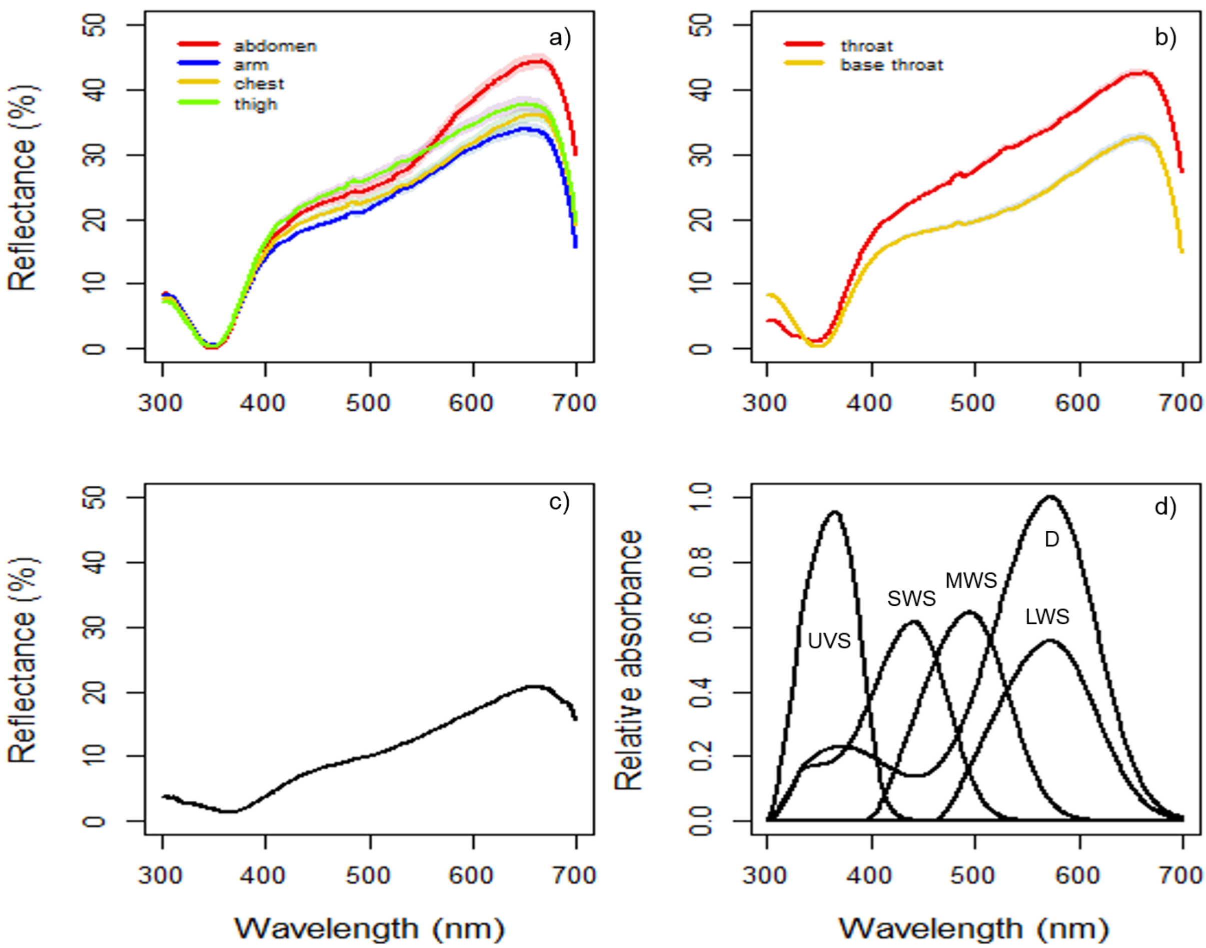
Table S9. The subset of 4 models with a ΔAICc < 4 relative to the best model among the 22 models considered to explain the probability of being infected by ticks in water dragons. 'K' denotes the number of parameters; 'AICc' is the AIC corrected for finite sample size; 'ΔAICc' is the difference between AICc of a given model to that of the best model; 'ωAICc' is the probability of each model given the data and the model set; 'logLik' is the log-likelihood; 'Cum. ωAICc' is the cumulative probability.

Model	K	AICc	ΔAICc	ωAICc	logLik	Cum. ωAICc
Chromatic base-throat contrast + PCA Chromatic	4	189.19	0	0.59	-90.24	0.59
PCA Chromatic	3	191.16	1.97	0.22	-92.37	0.81
Achromatic base-throat contrast + PCA Chromatic	4	192.72	3.53	0.10	-92.00	0.91
Achromatic throat contrast + PCA Chromatic	4	192.87	3.69	0.09	-92.08	1

Supplementary Figures

Figure S1. Adult male water dragons (*Intellagama lesueurii*) from Lane Cove National Park. The male on the left is an older male with a darker chest than the younger male on the right.

Figure S2. a) Abdomen, arm chest and thigh average (± standard errors) reflectance; b) Throat average (± standard errors) reflectance; c) Average background reflectance; d) Lizard spectral sensitivities (UVS: ultraviolet-sensitive; SWS: short wavelength sensitive; MWS: medium wavelength sensitive; LWS: long wavelength sensitive single cones; D: double cone; Barbour *et al.* 2002).



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